LAB ASSIGNMENT – 3

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Programs

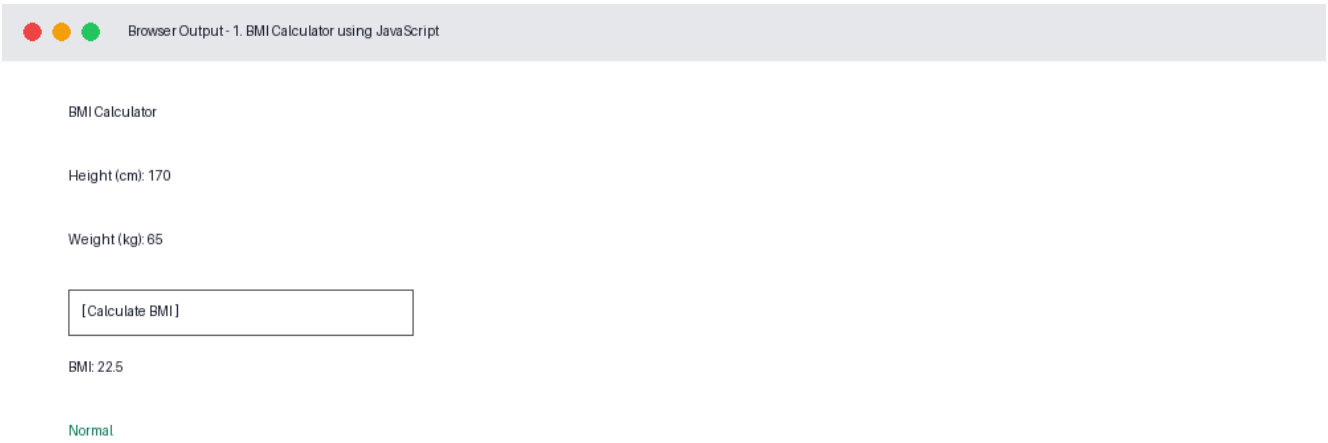
1. BMI Calculator using JavaScript
2. Signup Page with Form Validation
3. Change Color Theme
4. Digital Clock using JavaScript
5. To-Do List App

1. BMI Calculator using JavaScript

Code

```
<html>
<body>
<h2>BMI Calculator</h2>
Height (cm): <input type="number" id="h"><br><br>
Weight (kg): <input type="number" id="w"><br><br>
<button onclick="calculate()">Calculate BMI</button>
<p id="result"></p>
<script>
function calculate() {
    let h = document.getElementById("h").value / 100;
    let w = document.getElementById("w").value;
    if (h <= 0 || w <= 0) {
        document.getElementById("result").innerHTML = "Enter valid values";
        return;
    }
    let bmi = w / (h * h);
    if (bmi < 18.5) {
        document.getElementById("result").innerHTML = "BMI: " + bmi.toFixed(1) + "<br>Underweight";
    } else if (bmi < 25) {
        document.getElementById("result").innerHTML = "BMI: " + bmi.toFixed(1) + "<br>Normal";
    } else {
        document.getElementById("result").innerHTML = "BMI: " + bmi.toFixed(1) + "<br>Overweight";
    }
}
</script>
</body>
</html>
```

Output

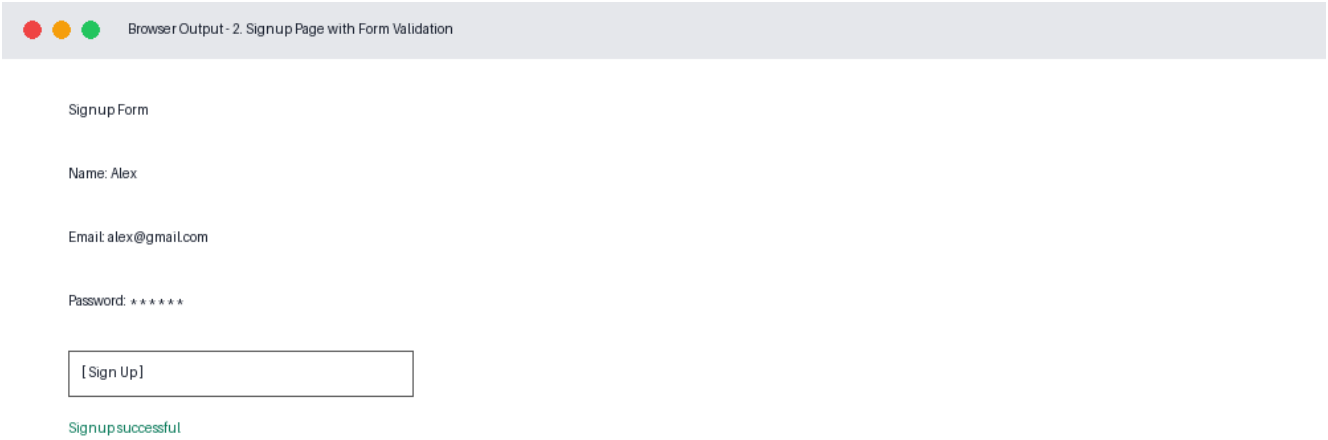


2. Signup Page with Form Validation

Code

```
<html>
<body>
<h2>Signup Form</h2>
Name: <input type="text" id="name"><br><br>
Email: <input type="text" id="email"><br><br>
Password: <input type="password" id="password"><br><br>
<button onclick="signup()">Sign Up</button>
<p id="message"></p>
<script>
function signup() {
    let name = document.getElementById("name").value;
    let email = document.getElementById("email").value;
    let password = document.getElementById("password").value;
    if (name == "" || email == "" || password == "") {
        document.getElementById("message").innerHTML = "Please fill all fields";
    } else if (!email.includes("@") || !email.includes(".")) {
        document.getElementById("message").innerHTML = "Enter valid email";
    } else if (password.length < 6) {
        document.getElementById("message").innerHTML = "Password must be at least 6 characters";
    } else {
        document.getElementById("message").innerHTML = "Signup successful";
    }
}
</script>
</body>
</html>
```

Output

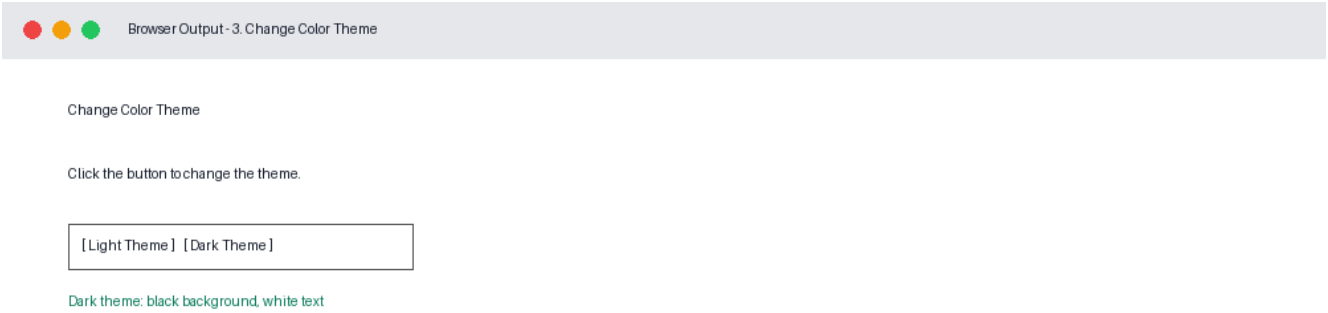


3. Change Color Theme

Code

```
<html>
<body id="page">
<h2>Change Color Theme</h2>
<p>Click the button to change the theme.</p>
<button onclick="light()">Light Theme</button>
<button onclick="dark()">Dark Theme</button>
<script>
function light() {
    document.getElementById("page").style.backgroundColor = "white";
    document.getElementById("page").style.color = "black";
}
function dark() {
    document.getElementById("page").style.backgroundColor = "black";
    document.getElementById("page").style.color = "white";
}
</script>
</body>
</html>
```

Output

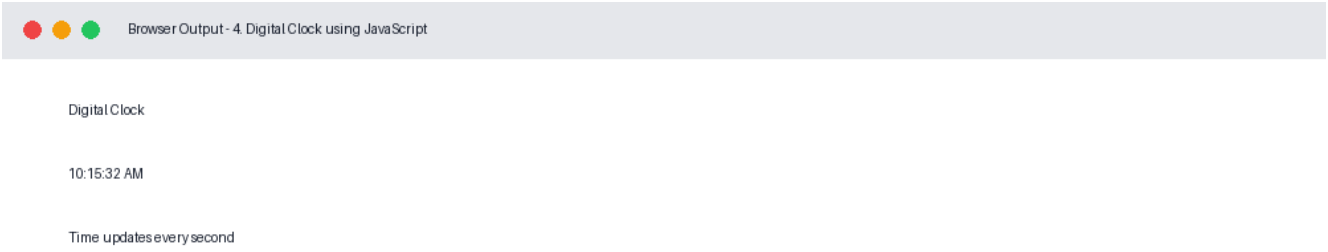


4. Digital Clock using JavaScript

Code

```
<html>
<body>
<h2>Digital Clock</h2>
<h1 id="clock"></h1>
<script>
function showTime() {
    let time = new Date();
    document.getElementById("clock").innerHTML = time.toLocaleTimeString();
}
showTime();
setInterval(showTime, 1000);
</script>
</body>
</html>
```

Output



5. To-Do List App

Code

```
<html>
<body>
<h2>To-Do List</h2>
<input type="text" id="task" placeholder="Enter a task">
<button onclick="addTask()">Add</button>
<ul id="list"></ul>
<script>
function addTask() {
    let task = document.getElementById("task").value;
    if (task.trim() == "") {
        alert("Enter a task");
        return;
    }
    let li = document.createElement("li");
    li.innerHTML = task + " ";
    li.onclick = function() {
        li.style.textDecoration = "line-through";
    };
    let btn = document.createElement("button");
    btn.innerHTML = "Delete";
    btn.onclick = function(event) {
        event.stopPropagation();
        li.remove();
    };
    li.appendChild(btn);
    document.getElementById("list").appendChild(li);
    document.getElementById("task").value = "";
}
</script>
</body>
</html>
```

Output

